Functional Pin Diagram & Signals of 8086

Q. Discuss the functional pin diagram of the 8086 microprocessor. Explain the significance of the multiplexed Address/Data bus and the pins specific to Minimum and Maximum modes.

Definition to Remember

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A uniqu
*Mode***

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1. Multiplexed Address/Data and Status Buses

* **AD0 - AD15 (Pins 16-2, 39):** Multiplexed Address/Data. During clock cycle T1, they carry the lower 16 bits of the memory address. During T2-T4, they carry 16-bit data.

* **A
address

2. Common Control & Interrupt Pins

* **ALE (Address Latch Enable):** Signals external latches to capture the address from the AD bus during T1.

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3. Minimum Mode Pins (MN/MX' tied to +5V)

Operates as a standalone processor generating its own control signals.

* **M/I

4. Maximum Mode Pins (MN/MX' tied to GND)

Used in multi-processor systems (e.g., with 8087 Math Coprocessor). Control signal generation is handed to an external 8288 Bus Controller.

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Must-Write Points (for 10 marks)

1. The 8

Quick Recall

`40-Pin
M/IO, H